

Donghwan Rho

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  Website
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Research Interests

- Implementing Language Models Under Homomorphic Encryption
- Mathematical Reasoning in Large Language Models
- Interpretability of Language Models

Education

- Ph.D. Seoul National University**, Mathematical Sciences Mar. 2021 – Present
- Advisor: [Ernest K. Ryu](#)
 - Expected Graduation: Feb. 2026
- BS Korea University**, Mathematics Education Mar. 2015 – Feb. 2021
- Graduated 1st in Class
 - (Mandatory) Military Service: May 2017 - Jan. 2019
 - Teacher's Certificate for Secondary Regular Teacher of Mathematics.

Experience

- Workshop Coordinator**, SNU-KAIST ML Theory Workshop Gangneung, South Korea Aug. 2024
- Assisted in planning the schedule, selecting the venue, and managing workshop programs.
- Teaching Practicum** Seoul, South Korea Apr. 2020 - May 2020
- Danggok high school.
- Military Service**, Auxiliary Police Seoul, South Korea May 2017 – Jan. 2019
- Assisted in maintaining public order and security.

Publications

*: Equal Contribution

- [2] **Homomorphically Encrypted Language Models Stabilized with TSP-Sorted Tokens**
Donghwan Rho*, Sieun Seo*, Hyewon Sung, Chohong Min, Ernest K. Ryu
 In Progress
- [1] **Encryption-Friendly LLM Architecture** Oct. 2024
Donghwan Rho*, Taeseong Kim*, Minje Park, Jung Woo Kim, Hyunsik Chae, Ernest K. Ryu,
 Jung Hee Cheon
 International Conference on Learning Representations (ICLR) 2025
[Paper](#) [Code](#)

Talks & Presentations

- SNU-Samsung Electronics Collaboration Symposium**, Poster Presentation Aug. 2024
- Title: Encryption-Friendly LLM Architecture
- SNU-KAIST ML Theory Workshop**, Poster Presentation Aug. 2024

- Title: Encryption-Friendly LLM Architecture

Honors and Awards

Outstanding TA , Seoul National University	July 2025
Introduction to Computer Programming and Artificial Intelligence for Scientists	
University Students Contest of Mathematics , Silver Prize	Dec. 2018
Academic Excellence Scholarship , Korea University	Aug. 2015

Teaching

Peer Tutoring in Basic Sciences , Lead Teaching Assistant	Mar. 2024 - Present
• Represented section TAs to ensure the smooth operation of the peer tutoring program. • Oversaw tutorings in Mathematics, Physics, Chemistry, Biology, and Statistics.	
Basic Calculus , Teaching Assistant	Sep. 2021 - Present
• Matched tutors and tutees for the <i>Basic Calculus</i> peer tutoring program. • Managed program logistics to ensure efficient operation.	
SNU Pre-College Program , Head TA, Instructor	
• Managed logistics of the mathematics program for freshman undergraduate. • Gave lectures. • Winter 2024, Winter 2023, Winter 2022	
Introduction to Computer Programming and Artificial Intelligence for Scientists , Instructor	Spring 2025
• English Course • Selected as an outstanding TA	
Topics in Machine Intelligence	Fall 2024
• Topics: Continuous-depth neural networks, Diffusion Models, Large Language Models, Vision Language Models, State-space Models • English Course • Graduate Course	
Mathematical algorithms II	Fall 2022
• Topics: Reinforcement Learning, Diffusion Models, Natural Language Processing • English Course • Graduate Course	
Calculus 2 Practice , Instructor	Fall 2021
Calculus 1 Practice , Instructor	
• Spring 2024, Fall 2023, Spring 2023, Spring 2022, Spring 2021	

Technical Skills

Languages: Korean (Native), English (Fluent)

Programming Languages: Python, C++

Tools & Technologies: LaTeX, GitHub